

Risk-controlled post-hoc rescue of rare cell types in single-cell RNA-seq under label scarcity

Anonymous Authors

Affiliation

corresponding@author

Abstract

Motivation: Rare cell types in single-cell RNA-seq are systematically missed by semi-supervised classifiers when few rare cells are labelled, and naively over-predicting the rare class inflates false positives that can be more damaging than the missed cells. A method recovering rare cells with an explicit error-rate bound is needed.

Results: scRareRefine is a post-hoc module refining a frozen scANVI backbone via an anisotropic prototype-membership score and a split-conformal threshold with three abstention gates, calibrated only on held-out data to target a false-rescue rate (FFR) budget of $\alpha = 0.01$ (empirical, not finite-sample). Across six datasets, nine methods and three splits, it raises mean rare-cell F1 in the label-scarce regime from 0.72 (scANVI) to 0.88 while keeping worst-case FFR at 0.0098, the only method achieving both. It abstains to the backbone when rescue is unwarranted, so it does not underperform scANVI (dataset-clustered bootstrap, sign-test $p = 0.031$). We characterise failure modes, including a recall ceiling for entangled rare types.

Availability and implementation: Code and reproduction scripts are available at <https://github.com/scrarerefine/scrarerefine>.

Contact: corresponding@author

1 Introduction

Single-cell RNA sequencing (scRNA-seq) routinely resolves the major cell types of a tissue, but the populations that matter most for a biological question are often the *rare* ones: a small dendritic-cell subset, a sparse endocrine population, or a handful of mast cells. Automated annotation methods trained with only a few labelled examples of such a type tend to absorb its cells into a transcriptionally adjacent majority class. Semi-supervised models such as scANVI (Xu et al., 2021), which combine a variational backbone with a small amount of label information, are a strong default for this task, yet they inherit the same difficulty: when the labelled budget for a rare type is small, the type is under-represented in the supervised signal and is systematically missed at test time.

A natural reflex is to make the classifier more willing to predict the rare class. This does recover some missed cells, but it also relabels majority cells that merely sit near the rare population, inflating false positives. In many downstream analyses a false rare call (e.g. reporting mast cells where there are none) is more costly than a missed one. The practical problem is therefore not simply “predict more rare cells” but “recover missed rare cells *without* paying an uncontrolled false-positive cost” — and to do so *inductively*, using only information available at training and calibration time.

We address this with **scRareRefine**, a lightweight post-hoc module that refines, rather than replaces, the predictions of a frozen scANVI backbone (Fig. 1). scRareRefine first builds a class prototype for every

labelled training type in the scANVI latent space and scores each query cell by an anisotropic membership distance to the rare prototype, normalised by each class’s compactness. It then decides which cells to relabel through a calibration step performed entirely on a held-out validation set: a conformal threshold on validation non-rare scores controls the false-rescue rate, and three abstention gates — a separability safety net, a necessity guard, and a validation-adaptive candidate-rank selector — prevent the module from acting when rescue is unsafe or unnecessary. The reference prototypes come only from the training set and every threshold comes only from the validation set; test labels are used exclusively for final evaluation.

The central design choice is to trade a little recall for a *controlled* false-rescue rate (FFR). We fix the FFR budget to a publication-grade $\alpha = 0.01$ across all datasets and never tune it, so that improvements in rare-cell F1 are never bought with an uncontrolled increase in false positives. Because the module abstains when its gates are not satisfied, it reduces exactly to the backbone on datasets where rescue is not warranted; its benefit is concentrated in the genuinely label-scarce, genuinely-missed regime.

Contributions.

1. **A risk-controlled rescue module.** scRareRefine is a fully inductive, post-hoc refinement of a frozen scANVI classifier that recovers missed rare cells while holding the false-rescue rate below a fixed budget $\alpha = 0.01$ (Section 2).
2. **Label-scarce gains with controlled FFR.** Across six datasets, nine methods and three splits, scRareRefine attains the highest mean rare-cell F1 in the label-scarce regime (0.88 vs. 0.72 for scANVI) and is the only method whose worst-case FFR stays $\leq \alpha$; it does not underperform scANVI in this regime, and a dataset-clustered bootstrap over the six datasets confirms the improvement is not driven by any single dataset (sign-test $p = 0.031$) (Section 3.2).
3. **Component analysis.** A leave-one-out ablation shows that the adaptive candidate-rank selector is the component that drives F1, the conformal threshold is what controls FFR, and the two gates act as safety mechanisms that prevent regressions rather than as performance boosters (Section 3.3).
4. **An honest analysis of the separability gate.** Through a controlled stress test and a benchmark-wide sensitivity sweep we show that the gate threshold is a deliberately conservative, pre-specified prior whose risk axis is dataset-dependent, and we report where it is too cautious (Section 3.4).
5. **Characterised failure modes.** We delineate the regime in which the method is a no-op and its concrete limitations, including a recall ceiling for rare types that are geometrically entangled with a majority population (Section 4).

Related work. *Reference-based annotation* methods — logistic-regression CellTypist (Domínguez Conde et al., 2022), imbalance-aware scBalance (Cheng et al., 2023), and the interpretable transformer TOSICA (Chen et al., 2023) — transfer labels from a reference but do not provide an explicit false-positive guarantee for a target rare type, and their accuracy degrades when the rare type is sparsely labelled. *Rare-population discovery* methods such as scCAD (Xu et al., 2024) detect anomalous clusters in an unsupervised manner; mapping such discoveries onto a *specific* rare type incurs an inherent precision penalty. *Prototype and semi-supervised* approaches, including the scVI/scANVI family (Lopez et al., 2018; Xu et al., 2021) and prototype classifiers, motivate our latent-space prototype scoring but do not target the scarce-label rescue problem with risk control. *Conformal prediction* (Vovk et al., 2005; Angelopoulos and Bates, 2023) provides the distribution-free machinery we use to translate a validation calibration set into a bound on the false-rescue rate. Conformal inference has very recently been introduced to single-cell annotation by López-De-Castro et al. (2025), who use it to attach coverage-guaranteed prediction sets and to flag out-of-distribution

scRareRefine

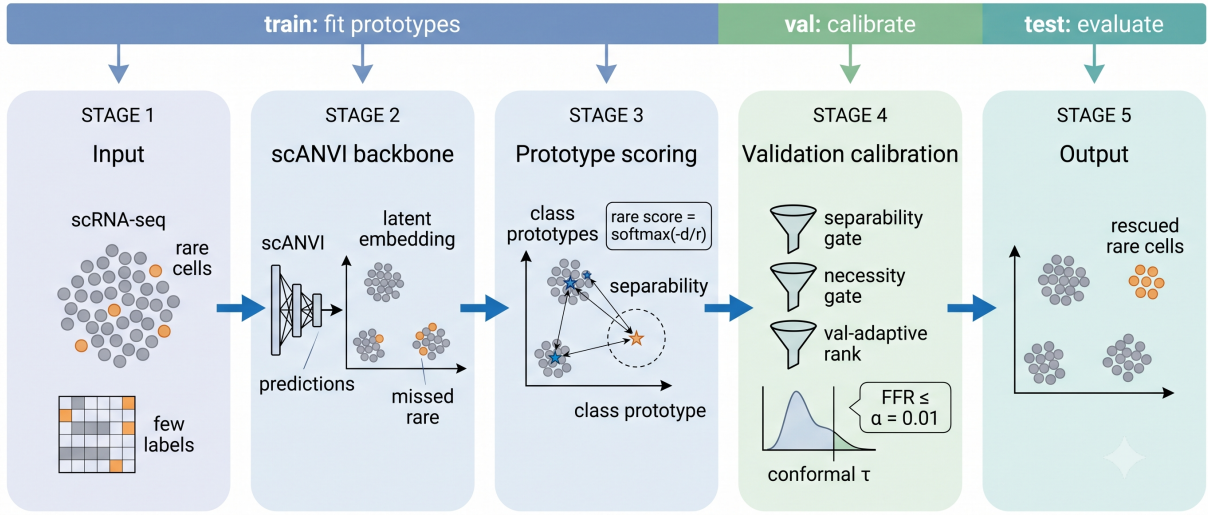


Figure 1: **Overview of scRareRefine.** A frozen scANVI backbone produces a latent embedding and a prediction for every cell. On the training set we build class prototypes; for each query cell we compute an anisotropic rare-membership score. A validation set calibrates a conformal threshold and three abstention gates so that the false-rescue rate stays below α . At test time, candidate cells whose score exceeds the threshold are relabelled to the rare class. Reference prototypes use only training cells; all thresholds use only validation cells.

cells; that work controls the *marginal coverage* of a prediction set over all cells, whereas scRareRefine uses conformal calibration for a different objective — a directed, per-type bound on the rate at which a specific rare type is falsely called. scRareRefine thus combines latent-space prototype geometry with conformal calibration to obtain a rescue that is both effective in the scarce regime and risk-controlled, a combination not offered by prior methods.

2 System and methods

2.1 Setup and inductive protocol

Let a dataset contain cells with expression profiles and a single designated rare type r . We split cells into training, validation and test partitions using a batch-/donor-held-out scheme (target 70/15/15), so that validation and test cells come from donors unseen during training; this is the harder, more realistic inductive setting. A semi-supervised scANVI model (Xu et al., 2021), itself initialised from an scVI model (Lopez et al., 2018), is trained on the training partition with all majority cells labelled and only a limited budget of rare cells labelled. The labelled rare budget is $\max(5, \lfloor p N_{\text{tr}}^r \rfloor)$ for a nominal fraction p , where N_{tr}^r is the number of rare cells in the training partition; the floor of five reflects the minimum needed to estimate a prototype. The model yields, for every cell, a latent embedding $z \in \mathbb{R}^d$ and a predicted label. scRareRefine consumes these frozen outputs and never retrain the backbone.

2.2 Prototype-membership scoring

For each class c observed among the *labelled* training cells we form a prototype $\mu_c = \frac{1}{|S_c|} \sum_{i \in S_c} z_i$, the mean latent of its labelled cells S_c , and a robust radius $r_c = \max(\text{median}_{i \in S_c} \|z_i - \mu_c\|, \epsilon)$ (set to a conservative default when $|S_c| < 3$). For a query cell with embedding z we define an *anisotropic rare-membership score*

$$s(z) = \text{softmax}_c \left(- \frac{\|z - \mu_c\|}{r_c} \right) [r], \quad (1)$$

i.e. the rare-class component of a softmax over per-class distances normalised by each class’s compactness. The per-class normalisation lets the score adapt to classes of different spread and limits intrusion from a tight neighbouring majority cluster. We also define a train-only *separability ratio*

$$\rho = \frac{\min_{c \neq r} \|\mu_r - \mu_c\|}{\frac{1}{|S_r|} \sum_{i \in S_r} \|z_i - \mu_r\|}, \quad (2)$$

the distance from the rare prototype to its nearest majority prototype divided by the mean rare intra-class radius; ρ measures how geometrically distinct the rare type is, using only training information.

2.3 Algorithm: conformal calibration and abstention gates

Given the frozen scores, scRareRefine decides which test cells to relabel using only validation labels. Three gates and one threshold are applied in sequence (Algorithm 1).

Separability gate. If $\rho < \rho_{\text{low}}$ the rare and majority populations are too entangled for prototype rescue to be safe and the module abstains. We fix $\rho_{\text{low}} = 1.3$ as a dataset-independent prior (Section 3.4).

Necessity guard. If the validation backbone already recovers the rare type — specifically if the number of validation rare cells it misses is below M — rescue is unnecessary and may only add noise, so the module abstains. We fix $M = 3$, the minimum support needed for the validation-adaptive selection below to carry signal.

Conformal threshold. Let the validation non-rare scores be $s_1, \dots, s_n$. We set the threshold τ to the finite-sample $(1 - \alpha)$ conservative order statistic $\tau = s_{(k^*)}$ with $k^* = \lceil (1 - \alpha)(n + 1) \rceil$ (and $\tau = +\infty$, i.e. no rescue, if $k^* > n$). Under exchangeability of validation and test non-rare cells, relabelling only cells with $s \geq \tau$ bounds the marginal probability that a non-rare cell is rescued at α .

Validation-adaptive candidate rank. A cell is a *candidate* only if the rare prototype is among its k nearest prototypes (isotropic Euclidean rank $\leq k$) and its backbone label is not already rare. We choose $k \in \{1, 2, 3\}$ on validation as the value that maximises validation rare-cell F1 subject to a Wilson 95% upper bound on the validation FFR not exceeding α , breaking ties toward the smaller (more conservative) k . Using the Wilson upper bound rather than a point estimate guards against the validation/test distribution shift induced by donor-held-out splits.

Rescue rule. A test cell is relabelled to r iff it is a candidate (rank $\leq k$) and $s \geq \tau$. All other predictions are left unchanged.

Algorithm 1 scRareRefine conformal rescue (inductive)

Require: frozen latents and predictions for train/val/test; rare type r ; budget α

- 1: fit prototypes $\{\mu_c\}$, radii $\{r_c\}$, separability ρ on labelled *train* cells
 - 2: **if** $\rho < \rho_{\text{low}}$ **then return** backbone predictions $\triangleright$ separability gate
 - 3: **end if**
 - 4: $v_{\text{miss}} \leftarrow \# \text{validation rare cells missed by backbone}$
 - 5: **if** $v_{\text{miss}} < M$ **then return** backbone predictions $\triangleright$ necessity guard
 - 6: **end if**
 - 7: compute scores Eq. (1) for val and test; set τ from val non-rare scores
 - 8: choose $k \in \{1, 2, 3\}$ maximising val rare-F1 s.t. Wilson-upper val-FFR $\leq \alpha$
 - 9: **for** each test cell: relabel to r if $\text{rank} \leq k$ and $s \geq \tau$
 - 10: **return** refined test predictions
-

2.4 False-rescue rate

We quantify the cost of rescue by the *false-rescue rate*,

$$\text{FFR} = \frac{|\{\text{non-rare test cells relabelled to } r\}|}{|\{\text{non-rare test cells}\}|}.$$

The conformal threshold and gates are designed so that FFR is controlled at the fixed budget $\alpha = 0.01$. Because validation and test cells come from different donors under our split, this control provides marginal (not per-dataset conditional) coverage and is empirical rather than a strict finite-sample guarantee; we report FFR with raw counts and treat α as a fixed prior, never tuned per dataset.

2.5 Implementation

scRareRefine is implemented as a lightweight post-processing layer in Python (NumPy/SciPy) that consumes only the frozen latent embeddings and label predictions exported by an existing scANVI/scVI model; it requires no modification to, or retraining of, the backbone, and adds negligible runtime (prototype fitting and scoring are $O(N_{\text{train}}d)$ and $O(N_{\text{test}}Cd)$ for C classes and latent dimension d). Benchmarking used scvi-tools for the scANVI/scVI backbone and standard scientific-Python tooling for prototypes, conformal calibration and evaluation; comparator methods used their respective published implementations. All splits, seeds, per-run manifests and the exact configuration used for every reported number are released with the code (Supplementary S1); see the Availability and implementation statement in the Abstract for the repository location.

3 Results

3.1 Experimental setup

We evaluate on six scRNA-seq datasets, each with a designated rare type: human immune dendritic cells (ASDC), Baron pancreas (gamma) (Baron et al., 2016), an integrated human pancreas (endothelial), and three Tabula Sapiens subsets (Tabula Sapiens Consortium, 2022) — lung (lymphatic endothelial), small intestine (tuft) and stomach (mast). For every dataset we use a donor-held-out 70/15/15 split and vary the labelled rare fraction over $\{0.01, 0.05, 0.10, \text{all}\}$, repeating each configuration over three random seeds (42, 43, 44). We compare scRareRefine against eight methods that read the same split and the same labelled subset: the scANVI backbone, a k -NN classifier on the scANVI latent, CellTypist, scBalance, ProtoCloud,

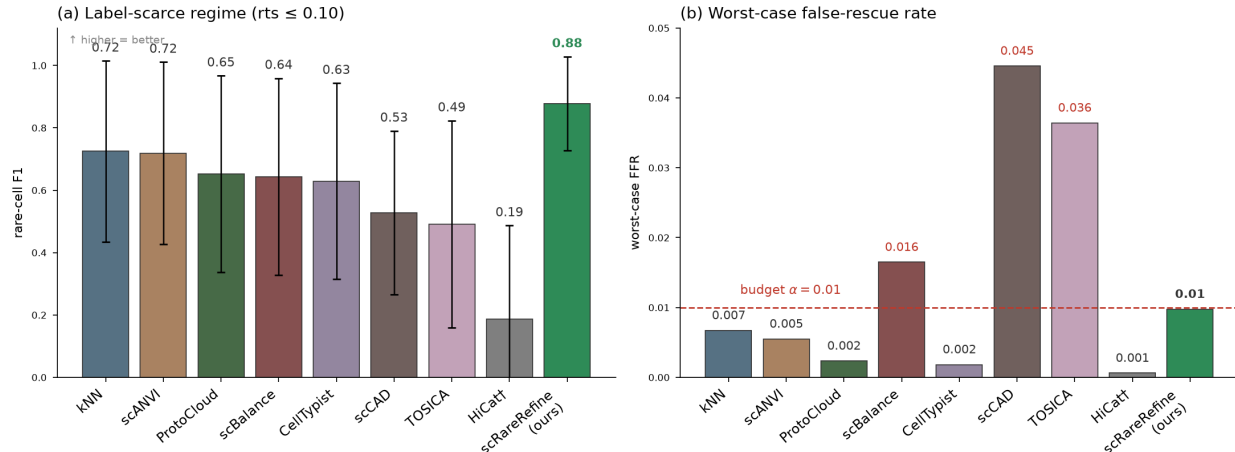


Figure 2: **Main result.** (a) Mean rare-cell F1 in the label-scarce regime ($p \leq 0.10$), averaged over six datasets, three fractions and three seeds. (b) Worst-case FFR over the whole benchmark; the dashed line is the budget $\alpha = 0.01$. scRareRefine is the only method that combines the highest scarce-regime F1 with a worst-case FFR $\leq \alpha$. † HiCat is transductive.

Table 1: **Per-dataset rare-cell F1 in the label-scarce regime** (mean over fractions $p \leq 0.10$ and three seeds). FFR_{max} is the worst-case false-rescue rate for scRareRefine over the same configurations.

Dataset (rare type)	scANVI	Best baseline	scRareRefine	FFR _{max}
immune dendritic (ASDC)	0.59	0.72 (kNN)	0.94	0.0010
Baron pancreas (gamma)	0.53	0.46 (kNN)	0.66	0.0098
lung (lymphatic EC)	0.72	0.92 (kNN)	0.98	0.0029
small intestine (tuft)	0.98	0.98 (kNN)	0.98	0.0005
stomach (mast)	0.61	0.49 (CellTypist)	0.72	0.0002
integrated pancreas (endo.)	0.89	0.98 (kNN)	1.00	0.0006

scCAD, TOSICA, and HiCat. HiCat is transductive (its dimensionality reduction is fit on combined train and test cells) and is therefore reported as an upper-bound reference rather than a fair inductive competitor; TOSICA is run with its default pathway configuration. We report rare-cell F1 and the false-rescue rate (FFR), aggregated as mean $\pm$ standard deviation over seeds. Because the labelled budget is $\max(5, \lfloor p N_{\text{tr}}^r \rfloor)$, several nominal fractions collapse to the same labelled set on small-rare datasets; we therefore report the label-scarce comparison over the 15 *distinct* labelled configurations rather than the 18 nominal ones.

3.2 scRareRefine improves rare-cell F1 in the scarce regime with controlled FFR

Across the label-scarce regime scRareRefine attains the highest mean rare-cell F1 of all nine methods (Fig. 2a): 0.88, against 0.72 for the scANVI backbone and 0.73 for the next-best inductive method (k -NN on the scANVI latent); the remaining methods range from 0.49 to 0.65, and the transductive HiCat reference is lower (0.19) because it returns no rare cells on several datasets. Per dataset (Table 1), scRareRefine is the best or tied-best inductive method on all six datasets, with the largest gains where the backbone is weakest (immune dendritic cells, $0.59 \rightarrow 0.94$; stomach mast cells, $0.61 \rightarrow 0.72$).

This F1 advantage is not bought with an uncontrolled false-positive cost. scRareRefine is the only method whose worst-case FFR over the whole benchmark stays within budget ($0.0098 \leq \alpha$, Fig. 2b); by contrast scCAD (0.045), scBalance (0.016) and TOSICA (0.036) exceed it. A paired analysis over the

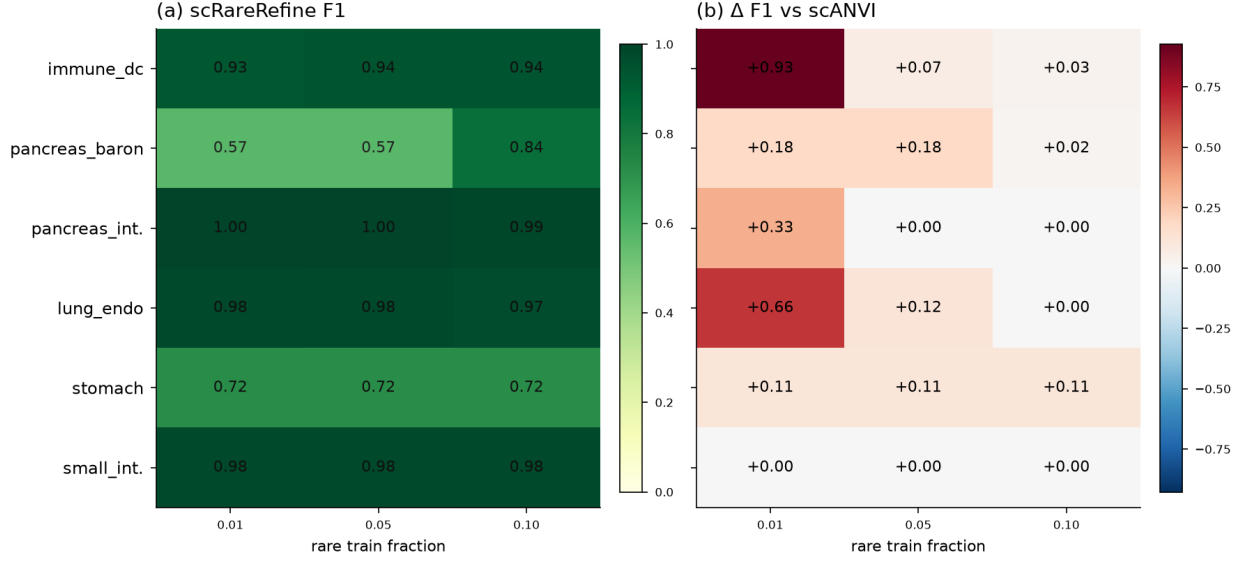


Figure 3: **Per-dataset rescue in the label-scarce regime** ($p \leq 0.10$, mean over three seeds). (a) scRareRefine rare-cell F1 for each dataset $\times$ labelled-fraction cell. (b) Improvement over the scANVI backbone ($\Delta F1 = \text{scRareRefine} - \text{scANVI}$); every cell is non-negative, so the module never lowers F1 relative to the backbone, and the largest gains fall on the sparsely labelled columns of the datasets the backbone handles worst (immune dendritic, lung, integrated pancreas at $p = 0.01$).

54 scarce configurations (six datasets $\times$ three distinct fractions $\times$ three seeds) shows that scRareRefine does not underperform the backbone: 29 wins, 25 ties — the ties are abstentions — and 0 losses, with a mean F1 improvement of $+0.16$ (per-configuration bootstrap 95% CI $[+0.09, +0.24]$, one-sided Wilcoxon $p \leq 1.3 \times 10^{-6}$ treating all 54 configurations as independent). Because seeds and collapsed fractions within a dataset are correlated, we treat that test as liberal and additionally run a *dataset-clustered* bootstrap that resamples whole datasets (the six biologically independent units) rather than configurations: against scANVI this gives a mean $\Delta F1$ of $+0.16$ with a 95% CI of $[+0.08, +0.25]$ (20 000 resamples) and, at the coarser per-dataset level, scRareRefine improves scANVI on 5/6 datasets with 0 losses (one exact tie, on the already-saturated small-intestine tuft dataset; sign-test $p = 0.031$). The two tests agree in direction and magnitude; we report both because the dataset-clustered version is the more conservative unit of resampling given the correlation structure (Supplementary S2). Because the labelled budget has a floor of five cells, the 18 nominal scarce configurations collapse to 15 distinct ones; over these, scRareRefine beats the majority of baselines in 15/15 and is the single best method in 14/15 (Supplementary S3).

Figure 3 resolves the per-method summary of Fig. 2 into the scRareRefine F1 grid and its improvement over the backbone: the gains concentrate in the most label-scarce columns of the datasets the backbone handles worst, and no cell is negative. Figure 4 shows the same effect as an across-configuration distribution — scRareRefine attains the highest and tightest F1 median (0.96) and lacks the long lower tail that the backbone and baselines inherit from the configurations on which they miss the rare type.

3.3 Component ablation

In the label-scarce regime the full method raises mean rare-cell F1 from 0.72 (backbone) to 0.88 (Fig. 5, left). Removing each component in turn separates two roles, and each ablated variant is itself a legitimate rescue baseline (an alternative post-hoc design one could have shipped instead of the full method), so we read this panel as a baseline comparison rather than only a component audit. The validation-adaptive candidate-rank

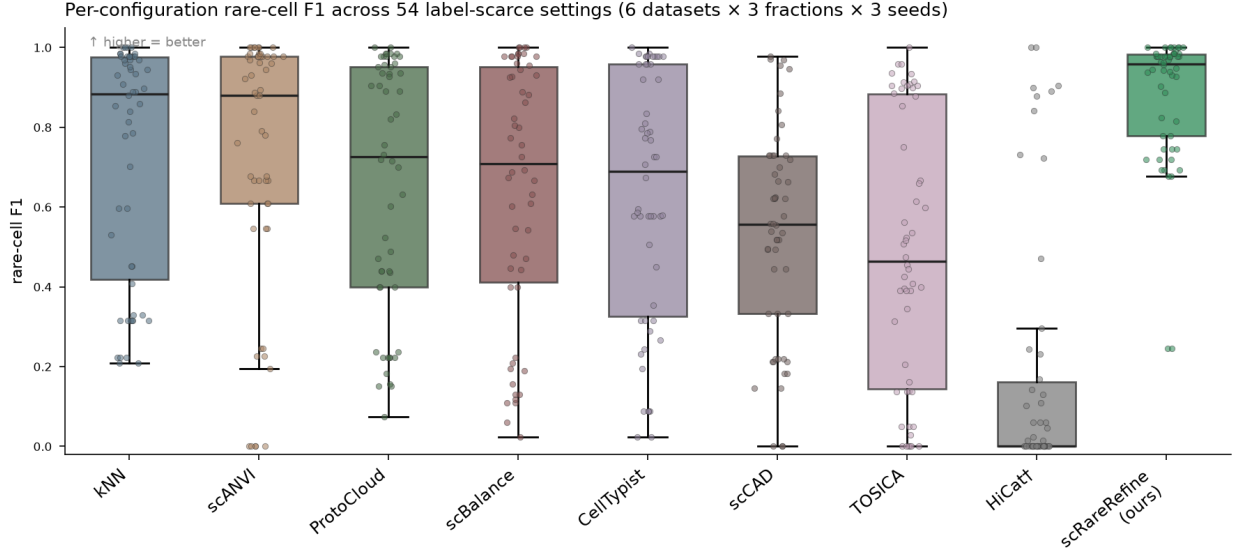


Figure 4: **Across-configuration distribution of rare-cell F1 in the label-scarce regime.** Each box summarises a method over 54 scarce configurations (six datasets $\times$ three fractions $\times$ three seeds); points are individual configurations. scRareRefine has the highest and tightest F1 distribution (median 0.96), while the backbone and the strongest baselines carry long lower tails from the datasets and fractions on which they miss the rare type. † HiCat is transductive.

selector is the only component that materially raises F1: replacing it with the most conservative fixed rank ($k = 1$) lowers mean F1 from 0.88 to 0.87. The conformal threshold and the two gates, by contrast, barely change mean F1 but are what keep the method safe. Removing the conformal threshold – an *uncalibrated-threshold rescue baseline* – leaves mean F1 essentially unchanged (0.88) yet pushes the worst-case FFR to 0.0165, above budget; removing the separability gate – an *ungated rescue baseline* that fires regardless of how entangled the rare type is with its neighbours – even *raises* mean F1 (to 0.90) but likewise breaks the budget (worst-case FFR 0.0153); and removing the necessity guard leaves the benchmark mean unchanged (0.88, worst-case FFR 0.0098) but causes per-dataset regressions on already-saturated datasets. Each of these three baselines is thus dominated by the full method once FFR is held to budget: they either match or exceed scRareRefine’s F1 but violate α , or match its FFR while regressing F1 on individual datasets – no ablated variant dominates the full method on both axes simultaneously. The rank-sensitivity panel (Fig. 5, right) confirms the value of adapting k : every fixed choice is dominated in F1 by the adaptive selector ($k=1$: 0.87, $k=2$: 0.85, $k=3$: 0.84 vs. adaptive 0.88), and the most permissive fixed rank ($k=3$) – a *fixed-rank-3 rescue baseline* – drives the worst-case FFR to 0.046, far above budget, whereas the adaptive selector stays within it (0.0098). In summary, the adaptive rank drives F1, the conformal threshold controls FFR, and the two gates prevent regressions rather than boosting F1; and against every one of these four ablated-variant baselines, the full method is the only configuration that stays within the FFR budget without giving up F1.

3.4 The separability gate is a conservative, dataset-dependent prior

The separability gate uses a fixed, dataset-independent threshold ($\rho_{\text{low}} = 1.3$). We probe this choice from two directions. First, a controlled semi-synthetic stress test progressively entangles a rare population (lung lymphatic endothelial cells) toward its nearest majority type; rescue *without* the gate stays within the FFR budget down to separability ≈ 0.76 and produces a single marginal violation (18 false rescues out of 1716,

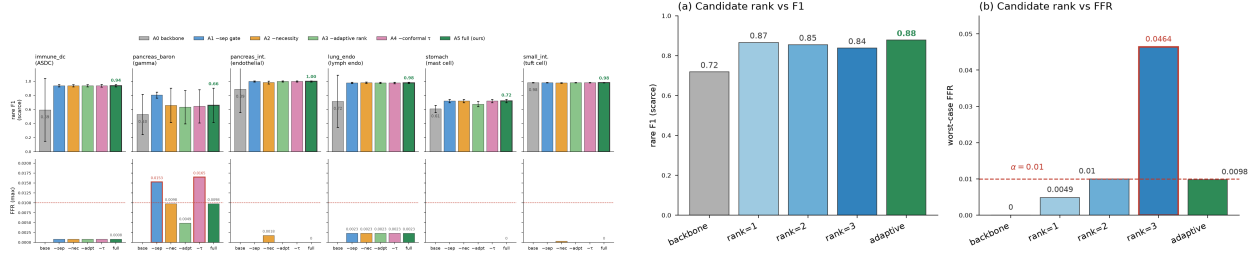


Figure 5: **Ablation (label-scarce regime), read as a baseline comparison.** Each leave-one-out variant is an alternative rescue design (an uncalibrated-threshold, ungated, or fixed-rank baseline) rather than only a component audit. Left: the four leave-one-out variants, per dataset (F1 top row, worst-case FFR bottom row); the dashed line is the budget $\alpha = 0.01$. Right: candidate-rank sensitivity (fixed $k \in \{1, 2, 3\}$ baselines vs. the validation-adaptive selector), mean F1 and worst-case FFR. Removing the conformal threshold or the separability gate barely changes — or even raises — mean F1 but breaks the FFR budget; the adaptive rank is the component that raises F1; a fixed rank = 3 over-fires FFR. No ablated-variant baseline dominates the full method on both F1 and FFR simultaneously.

FFR 0.0105) only at the lowest separability (≈ 0.69), well below the gate (Fig. 6). Second, a benchmark-wide sweep of the gate threshold (Supplementary S4) shows that 1.3 is the smallest threshold that keeps the worst-case FFR within budget on real data: lowering it to ≤ 1.0 raises mean scarce-regime F1 by 0.024 ($0.878 \rightarrow 0.902$) but admits two FFR violations (worst-case 0.0153, on the low-separability Baron-pancreas configurations at $\rho \approx 1.22$), whereas raising it to 1.6 only sacrifices F1 ($0.878 \rightarrow 0.838$) with no further FFR benefit. The two experiments disagree on where rescue becomes unsafe (≈ 1.22 on real Baron pancreas vs. ≈ 0.69 on the synthetic sweep), which is the central observation: separability alone does not monotonically determine rescue risk. We therefore present 1.3 not as a located collapse boundary but as a deliberately conservative, pre-specified prior chosen by the FFR constraint rather than tuned for F1 — lowering it would in fact *raise* mean F1 — that occasionally abstains where rescue would have been safe.

3.5 Qualitative rescue in latent space

We visualise the scANVI latent embedding with UMAP (McInnes et al., 2018).

Fig. 7 visualises the rescue on immune dendritic cells in the scANVI latent space. Of the true rare cells, the backbone recovers those forming a compact cluster (recall 0.72). scRareRefine relabels 23 further cells lying at the edge of this cluster, raising recall to 0.90 at precision 0.99, with a single false rescue. The 13 rare cells that remain missed are scattered far from the rare prototype; admitting them would require widening the candidate pool past the FFR budget. This is the geometric recall ceiling discussed in Section 4.

4 Discussion

scRareRefine is a conservative, risk-controlled rescuer, and its value is concentrated in a specific regime rather than uniform across all settings. We make this scope explicit.

The method is conditional by design. Two abstention gates cause scRareRefine to return the unmodified backbone prediction whenever the rare and majority prototypes are too entangled or the validation backbone already recovers the rare type. On datasets where the backbone is already saturated, the module therefore reduces exactly to scANVI — it neither helps nor harms — and its benefit is restricted to the genuinely label-scarce, genuinely missed regime. This conditional behaviour is what underwrites the “never underperforms

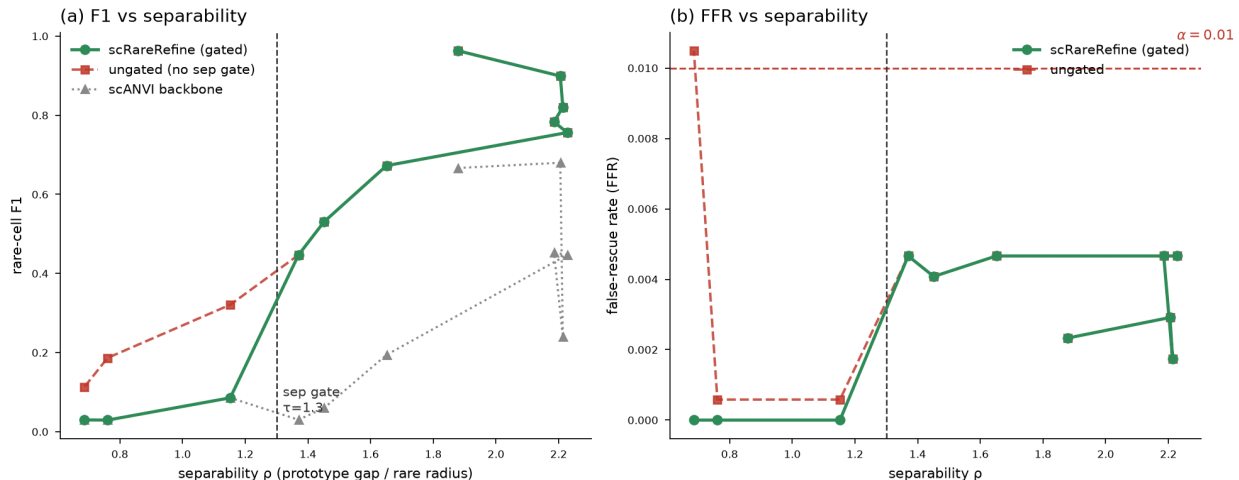


Figure 6: **Separability-gate analysis.** Controlled semi-synthetic stress test that progressively entangles a rare type toward its nearest majority population, read from right to left as entanglement increases (the x-axis shows separability ρ on its natural, increasing scale). Ungated rescue stays FFR-safe well below the gate at $\rho = 1.3$ and produces only a single marginal violation at the lowest separability, on the left edge of the plot; the gate abstains ($\rho < 1.3$) before rescue reaches that point. The budget $\alpha = 0.01$ is the dashed line.

the backbone” property: it holds by construction of the abstention mechanism, so the paired comparison should be read as evidence that rescue, when it fires, does not hurt, rather than as an unconditional dominance claim.

A recall ceiling for entangled rare types. For a rare type that is transcriptionally interleaved with a majority population, prototype-distance rescue has an intrinsic ceiling: a substantial fraction of true rare cells fall at candidate rank ≥ 3 , where admitting them would require widening the candidate pool past the point at which the FFR budget is exceeded. The adaptive-rank selector correctly refuses to do so, so recall plateaus (Fig. 7). Breaking this ceiling likely requires expression-side signals (e.g. marker genes) and is left to future work; we report below that a first attempt at incorporating such signals did not succeed under our inductive constraints, so this is a harder problem than it may appear.

Marker-gene validation did not help. Before converging on the purely embedding-based prototype-and-conformal design used in the main pipeline, we implemented and evaluated two earlier strategies that additionally scored rescue candidates against curated marker-gene expression: a marker-gated variant (`gate_marker`) and a linear score-fusion variant (`fusion`) that combined the prototype-membership score with a marker-expression score. On our development data, all three strategies – prototype-only gating, marker-gated, and fusion – converged to the *same* rare-cell F1, and the marker signal changed neither the selected candidates nor the outcome. We also found that the natural marker-threshold analogue of our conformal calibration is not informative in this setting: the validation-calibrated marker threshold collapses to the least restrictive value the search considers (equivalent to no filtering), and the corresponding test-set false-rescue rate at that threshold is zero, which reflects an already-clean candidate set from the embedding geometry rather than any discriminative contribution from marker expression. We disclose this negative result rather than omitting it: it indicates that, at least for the rare-vs-majority pairs in our six datasets, curated marker panels are largely redundant with the scANVI latent geometry once anisotropic prototype distance is used, and that closing the recall ceiling above will likely require a different source of signal (e.g. pathway-

level scores, spatial context, or a learned rather than curated marker set) rather than a simple threshold or linear fusion on top of the existing markers. This experiment used only training/validation partitions and is consistent with the inductive, test-blind protocol used throughout; no marker-based mechanism is part of the released method.

FFR control is marginal and empirical, and can sit near the boundary. The conformal threshold provides marginal FFR control under validation calibration, not a per-dataset conditional or strict finite-sample guarantee on test: under donor-held-out splits the exchangeability assumption is only approximately satisfied, and on one dataset (Baron pancreas) the worst-case test FFR reaches the $\alpha = 0.01$ budget. That worst case rests on the dataset with the fewest rare cells and thus the widest binomial uncertainty, so the true operating point could sit on either side of the budget; we mitigate this with a Wilson upper bound for rank selection and report all FFR values with raw counts, but the budget should be read as empirical, marginal control under distribution shift rather than a guarantee.

Seed sensitivity at extreme scarcity. At the smallest budget (five labelled rare cells) the improvement on one dataset is positive on average but unstable across seeds, mirroring the variance of the backbone itself; the instability disappears once at least ten rare cells are labelled. We therefore do not claim stable improvement at the five-label point on that dataset.

Evaluation caveats. Because the labelled budget has a floor of five, several nominal label fractions collapse to the same labelled set on small-rare datasets; we report distinct configurations to avoid double-counting. The per-configuration significance test pairs predictions by (dataset, fraction, seed); because these cells are correlated across seeds and near-duplicated across collapsed fractions, we treat that test as liberal and use a dataset-clustered bootstrap (resampling whole datasets) as our primary evidence instead (Supplementary S2). With only six datasets, the dataset-clustered analysis has limited power in absolute terms – the best it can report is a sign-test $p = 0.031$ – and we report it precisely because it is the honest, conservative unit of replication rather than because it is a strong result on its own. We evaluate six predominantly human single-cell datasets and do not test cross-species or disease-state transfer, both of which would further stress the exchangeability assumption. Finally, scRareRefine operates on scANVI embeddings; whether the same conformal rescue adds value on top of single-cell foundation-model embeddings is untested.

Relation to conformal single-cell annotation. López-De-Castro et al. (2025) bring conformal prediction to single-cell annotation to produce prediction sets with guaranteed marginal coverage and to detect out-of-distribution cells. scRareRefine shares the conformal calibration machinery but targets a different quantity: rather than covering the true label of every cell, it bounds the rate at which one designated rare type is falsely assigned. The two are complementary — a coverage-guaranteed prediction set quantifies uncertainty for all cells, whereas a directed false-rescue bound is what a practitioner needs when the decision is specifically whether to trust newly reported rare cells.

Outlook. The combination of latent-space prototype geometry with conformal calibration offers a general template for risk-controlled post-hoc refinement. Natural extensions include expression-side scoring to address the recall ceiling, batch-conditional calibration to tighten FFR control under donor shift, and application on top of foundation-model embeddings.

5 Conclusion

We introduced scRareRefine, a fully inductive post-hoc module that refines a frozen scANVI classifier to recover rare cell types under label scarcity while keeping the false-rescue rate below a fixed budget. By scoring cells with training-only prototype distances and gating relabelling with validation-only conformal calibration, the method raises rare-cell F1 in the label-scarce regime — the only method in our benchmark to do so with a controlled worst-case false-rescue rate — and abstains when rescue is not warranted, a property that holds by construction rather than as an unconditional dominance claim (§4, “The method is conditional by design”). We further provided an honest analysis of its conservative separability gate, its failure modes, and a prior internal negative result on marker-gene fusion. Risk-controlled refinement is a practical way to recover rare populations without paying an uncontrolled false-positive cost, and we hope it encourages reporting the operating point, not only the accuracy, of rare-cell annotation methods.

In summary: (i) scRareRefine is a post-hoc, inductive module that rescues missed rare cells on top of a frozen scANVI classifier without retraining the backbone; (ii) it couples training-only prototype-membership scoring with validation-only conformal calibration and three abstention gates to hold the false-rescue rate below a fixed budget $\alpha = 0.01$; (iii) in the label-scarce regime it raises mean rare-cell F1 from 0.72 to 0.88 across six datasets while being the only method with worst-case FFR $\leq \alpha$, and it abstains – returning the backbone – when rescue is unsafe; and (iv) the leave-one-out baseline comparison attributes F1 gains to the validation-adaptive candidate rank and FFR control to the conformal threshold and gates, while a controlled separability sweep shows the gate is a deliberately conservative, not an over-fitted, prior.

Supplementary material

Supplementary Material (S1–S8: reproduction protocol, dataset-clustered and per-configuration significance tables, distinct labelled configurations, and the separability-gate threshold sweep) is provided as a standalone document, `supplement.pdf`, distributed alongside this manuscript and referenced throughout the text as “Supplementary S_n ”.

Acknowledgements

We thank the maintainers of the public single-cell atlases used in this study.

Funding

[To be completed.]

References

- Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/22000000101.
- Maayan Baron, Adrian Veres, Samuel L Wolock, Aubrey L Faust, Renaud Gaujoux, Amedeo Vetere, et al. A single-cell transcriptomic map of the human and mouse pancreas reveals inter- and intra-cell population structure. *Cell Systems*, 3(4):346–360, 2016. doi: 10.1016/j.cels.2016.08.011.

- Jiawei Chen, Hao Xu, Wanyu Tao, Zhaoxiong Chen, Yuxuan Zhao, and Jing-Dong J Han. Transformer for one stop interpretable cell type annotation. *Nature Communications*, 14(1):223, 2023. doi: 10.1038/s41467-023-35923-4.
- Yuqi Cheng, Xingyu Fan, Jianing Zhang, and Yu Li. A scalable sparse neural network framework for rare cell type annotation of single-cell transcriptome data. *Communications Biology*, 6(1):545, 2023. doi: 10.1038/s42003-023-04928-6.
- C Domínguez Conde, C Xu, L B Jarvis, D B Rainbow, S B Wells, T Gomes, et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*, 376(6594):eabl5197, 2022. doi: 10.1126/science.abl5197.
- Romain Lopez, Jeffrey Regier, Michael B Cole, Michael I Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.
- Marcos López-De-Castro, Alberto García-Galindo, José González-Gomariz, and Rubén Armañanzas. Conformal inference for reliable single cell RNA-seq annotation. *Bioinformatics*, 41(10):btaf521, 2025. doi: 10.1093/bioinformatics/btaf521.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. doi: 10.48550/arXiv.1802.03426.
- Tabula Sapiens Consortium. The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896, 2022. doi: 10.1126/science.abl4896.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.
- Chenling Xu, Romain Lopez, Edouard Mehlman, Jeffrey Regier, Michael I Jordan, and Nir Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17(1):e9620, 2021. doi: 10.15252/msb.20209620.
- Yunpei Xu, Shaokai Wang, Qilong Feng, Jiazhi Xia, Yaohang Li, Hong-Dong Li, and Jianxin Wang. scCAD: Cluster decomposition-based anomaly detection for rare cell identification in single-cell expression data. *Nature Communications*, 15(1):7561, 2024. doi: 10.1038/s41467-024-51891-9.

scRareRefine rescue on immune_dc (ASDC, seed=42, 13 labeled rare, sep=2.14)
scANVI recall 0.72 → scRareRefine recall 0.90, precision 0.99

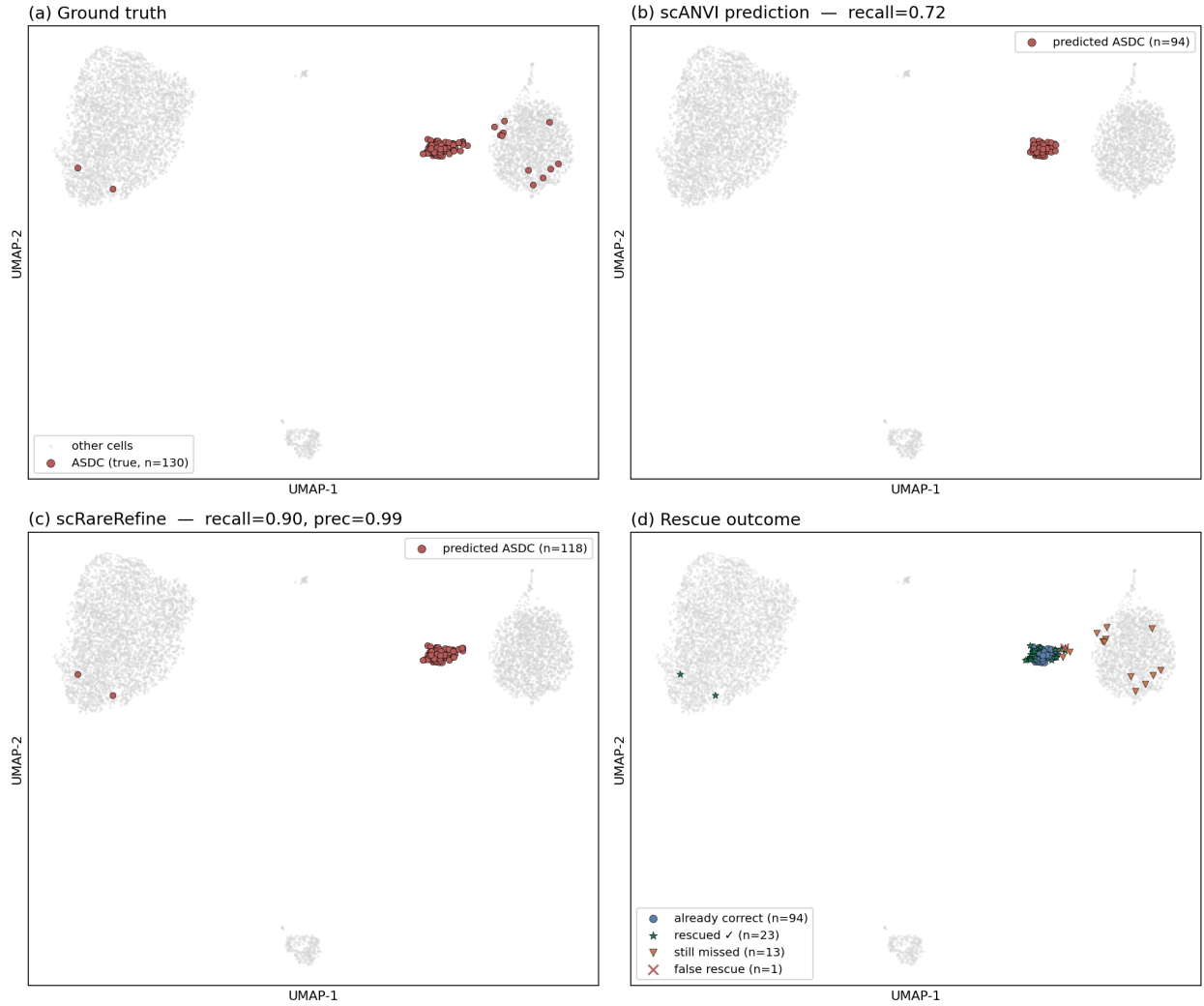


Figure 7: **Rescue in the scANVI latent space (immune dendritic cells).** Cells missed by scANVI that lie near the rare cluster are recovered (recall 0.72 → 0.90), whereas geometrically distant rare cells remain missed, illustrating the recall ceiling discussed in Section 4.